

Step-by-step guide: Attaching an RJ45 connector to a CAT6 cable

Tools and materials

- CAT6 Ethernet cable
- RJ45 connector (compatible with CAT6 cable)
- RJ45 crimping tool
- Cable stripper or stripping blade
- Wire cutters (if not built into the crimper)
- Network cable tester (recommended)

1. Strip the cable jacket

Use the cable stripper to remove about 2–3 cm (around 1 inch) of the outer jacket from the end of the CAT6 cable. Be careful not to nick or cut the insulated conductors inside.

2. Separate and straighten the wires

Untwist the four wire pairs and straighten each conductor with your fingers. Keep the untwisted section as short as possible to help maintain cable performance.

3. Arrange the wires

Choose a wiring standard. The most common is **T568B**, with the wires arranged from left to right as:

1. White/Orange
2. Orange
3. White/Green
4. Blue
5. White/Blue

6. Green
7. White/Brown
8. Brown

Ensure the wires are flat, straight, and in the correct order.

4. Trim the wires evenly

Hold the wires firmly and trim their ends so they are all the same length, leaving enough exposed conductor to reach the contacts inside the RJ45 plug.

5. Insert the wires into the RJ45 connector

With the connector's clip facing downward (or as required for your orientation), carefully slide the wires into the connector. Check that:

- Every wire reaches the end of the connector.
- Each wire stays in its correct slot.
- The cable's outer jacket extends slightly into the connector so it can be secured during crimping.

6. Crimp the connector

Insert the RJ45 connector fully into the crimping tool and squeeze the handles firmly until the crimping cycle is complete. This pushes the metal contacts into the conductors and locks the connector onto the cable jacket.

7. Inspect your work

Visually verify that:

- All eight wires are still in the correct order.
- The contacts have pierced each conductor.
- The cable jacket is held securely by the connector's strain relief.

8. Test the cable

Use a network cable tester to confirm that all eight conductors are connected correctly and that there are no opens, shorts, or crossed wires before using the cable.